Low-Frequency Low-Impedance Electromagnetic Shielding

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Abstract—**The problem of low-frequency shielding of a loop axially perpendicular to a plane shield of infinite extent is analyzed by 1) the thin shield work of S. Levy, 2) solution of the vector wave equation, and 3) application of the Schelkunoff transmission theory of shielding.**

Experimental data are obtained and compared with results of parts 1) and 3) in the frequency range 100–50 kHz. The first analytical technique is not general, and the limits of applicability of the results are discussed.

In the second solution, which is general, expressions are derived for the total electric and magnetic fields on both sides of and within the shield. The resulting expression for shielding effectiveness has been shown to agree quite accurately with experimental data. The final expression for shielding effectiveness shows *independence* of shielding with source-to-shield spacing, a result which is shown to be consistent with experimentally obtained data.

The results of the third theory are adapted to the problem. The shielding effectiveness expression $S = R + A + B$ is computer evaluated. It is shown that the formulations of Schelkunoff for shielding effectiveness are most accurate if and only if the shield is placed 1/3 of the distance from the source to the point of observation for calculation purposes.

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I. Introduction

THERE ARE numerous analyses used in the solution of shielding problems. Three analytical techniques are considered here in solving a shielding problem of fundamental importance: low-frequency electromagnetic shielding of a circular loop field source by a thin plane metal sheet of infinite extent. Results should be extendable to practical problems having more complex geometries. The first technique discussed is a theoretical determination of the fields by considering the induced eddy currents [1]. The second method is an exact solution of the vector wave equation [2], [3], [15]. The third method is an extension of the transmission theory of shielding [4], [5]. Each analytical method has advantages and disadvantages for different classes of shielding problems. The first and second methods are easily applied only in the limited class of problems where the geometry of the problem fits one of the natural coordinate systems. Cylindrical coordinates will be applicable to the problem at hand. The third method is of great value to the large class of plane wave shielding problems. An adaptation of this theory to the present problem is included [5]. It is shown that the adaptation incorrectly includes source-to-shield distance as a reflection loss parameter. Experimentally obtained data are used to check the results using both the Levy and the Schelkunoff methods.

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Throughout this study, unless otherwise noted, shielding effectiveness of a particular shield will be defined as the flux density or voltage insertion loss, in decibels, created by the introduction of the shield between a loop-field source and a loop-field sensor.

II. Theory

The first analytical method to be discussed is based on the classical eddy-current thin shield work of Maxwell. Levy adapted this analysis to this problem in 1933. He formulated the magnetic potentials and satisfied boundary conditions by postulating moving images. A discussion of his results follows. See Fig. 1. The total voltage induced in loop 1 due to eddy currents in the shield is

$$V_1' = M(2r_1) I\omega \cos \omega t + \int_{2r_1}^{\infty} \frac{dM(z)}{dz} I\omega \cos \omega \times \left(t - \frac{z - 2r_1}{V}\right) dz. \quad (1)$$

To expedite integration, Levy then formed an approximate equation[1] for $M(z)$

$$M(z) = 28.6n^2 e^{-9z/5a} 10^{-9} \text{ henrys}. \quad (2)$$

This approximation will limit the solution to ten-percent accuracy within the limits $0.15 < r_1/a < 0.9$. Other approximate forms of $M(z)$ could be derived from existing tables to extend the range of applicability of this solution.

Making this substitution in V_1' of (1) and performing the integration, Levy obtained

$$V_1' = -I(R' \sin \omega t + L'\omega \cos \omega t) \quad (3)$$

where

$$R' = \frac{1.42n^2 10^{-5} (1/\sigma_r, t) e^{-3.6r_1/a}}{1 + (78.8/at\sigma_r f)^2} \quad (4)$$

and

$$L' = \frac{-28.6n^2 10^{-9} e^{-3.6r_1/a}}{1 + (78.8/at\sigma_r f)^2} \quad (5)$$

where

R', L' are the apparent changes in real and imaginary parts of the impedance of loop 1

[1] See [6] and [7] in the bibliography of the paper by Levy[1].

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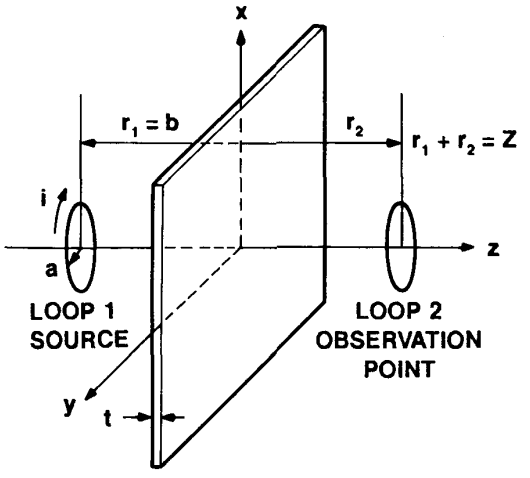


Fig. 1. Coaxial loops separated by an infinite plate. (Loop radius = 3.5 cm).

- n is the number of turns in loop 1
- σ_r is the percent conductivity of the shield material
- t is the shield thickness (in centimeters)
- r_1 is the distance in centimeters from loop 1 to shield (see Fig. 1)
- a is the radius of loop 1 (in centimeters)
- f is the frequency (in hertz).

To solve the actual shielding problem, Levy assumed the eddy currents to be distributed such that the induced potential on both sides of the shield would be identical. (Note that this would seem to restrict validity of results to shields of thickness less than a skin depth.) With this assumption, and by replacing $2r_1$ by $r_1 + r_2$ and n^2 by nn' in (3), one obtains, for the voltage induced in loop 2

$$V_2' = \frac{-1.42(1/\sigma_r t)10^4 - j28.6\omega}{1(78.8/at\sigma_r f)^2} nn' 10^{-9} e^{-1.8(r_1+r_2)/a} I.$$

It is important to note that V_2' is independent of r_1 or r_2 taken separately; that is, V_2' will be independent of the position of the shield between the two loops as long as the sum $r_1 + r_2$ is unchanged.

With the shield removed, the voltage in loop 2 is

$$V_2'' = -j\omega M_{12} I = -j\omega 28.6ann' 10^{-9} Ie^{-9(r_1+r_2)/5a}.$$

Adding the two opposing phasor voltages V_2' and V_2'' to obtain the total voltage induced in loop 2 yields

$$V_2 = V_2' + V_2''. \tag{6}$$

Then the insertion loss of the shield is

$$\left|\frac{V_2''}{V_2}\right| = \left|\frac{28.6\omega a[1+(78.8/at\sigma_r f)^2]}{-1.42(1/\sigma_r t)10^4 - j(78.8/at\sigma_r f)^2 \omega 28.6a}\right|$$

$$= [(at\sigma_r f/78.8)^2 + 1]^{1/2}.$$

The voltage drop (in decibels) caused by introducing the shield between the two loops is therefore

$$S \equiv 20 \log_{10}\left|\frac{V_2''}{V_2}\right| = 10 \log_{10}[(at\sigma_r f/78.8)^2 + 1] \tag{7}$$

where the symbols have the same meaning as in (4) and (5), and the accuracy is still dependent on the approximations made in forming $M(z)$.

Note that S as given here is apparently independent of the loop spacing z and the permeability of the metal plate. The experimental work described later shows some moderate variation of S with z. (See Figs. 11–13 later.) Thus, within the limits $0.15 < r_1/a < 0.9$ for ten-percent accuracy, the omission of z in (7) seems reasonable. Figs. 2–4 are graphs of (7) for the two thicknesses of aluminum, copper, and steel used in the experimental work to follow. The experimental curves are those corresponding to loop spacing $z = 3$ cm. (See Fig. 10 later.)

Bannister has shown by numerical analysis [13] that (7) follows from (54) and (55) developed herein.

Attention now turns to a solution of the problem by solving the vector wave equation for the vector potential:[2]

$$\nabla^2 \vec{A} + k_0^2 \vec{A} = 0. \tag{8}$$

The geometry of the problem suggests that the work might be done in cylindrical coordinates of unit vectors $\hat{l}_r$, $\hat{l}_\phi$, and $\hat{l}_z$, as shown in Fig. 5.

$$\vec{A} = A_\phi \hat{l}_\phi$$

$$\nabla \cdot \vec{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}. \tag{9}$$

But

$$A_r = 0 = A_z; \quad \frac{\partial}{\partial \phi} = 0 \text{ (cylindrical symmetry)} \tag{10}$$

therefore

$$\nabla \cdot \vec{A} = 0. \tag{11}$$

Similarly

$$\nabla \times \vec{A} = -\hat{l}_r \frac{\partial A_\phi}{\partial z} + \hat{l}_z \left\{\frac{\partial A_\phi}{\partial r} + \frac{1}{r} A_\phi\right\}. \tag{12}$$

Performing the curl again, we have

$$\nabla \times \nabla \times \vec{A} = -\hat{l}_\phi \left\{\frac{\partial^2 A_\phi}{\partial z^2} + \frac{\partial^2 A_\phi}{\partial r^2} + \frac{1}{r}\frac{\partial A_\phi}{\partial r} - \frac{1}{r^2} A_\phi\right\}. \tag{13}$$

Using the identity

$$\nabla^2 \vec{A} = \nabla(\nabla \cdot \vec{A}) - \nabla \times \nabla \times \vec{A}$$

the wave equation becomes

$$\frac{\partial^2 A_\phi}{\partial z^2} + k_0^2 A_\phi = \frac{1}{r^2} A_\phi - \frac{1}{r}\frac{\partial A_\phi}{\partial r} - \frac{\partial^2 A_\phi}{\partial r^2}.$$

[2] This problem has also been considered by R. K. R. Jonnada of the Moore School of Electrical Engineering, University of Pennsylvania, Philadelphia, PA.

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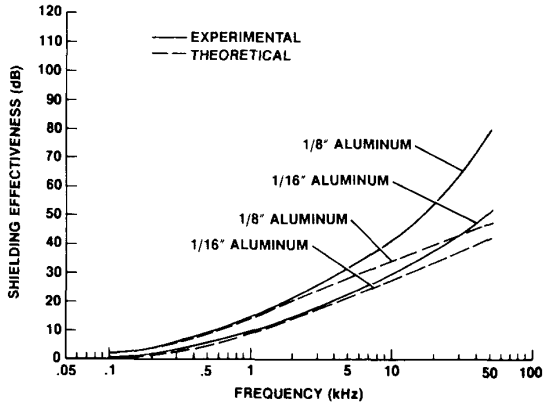


Fig. 2. Comparison of shielding effectiveness S for aluminum by two techniques (Loop radius = 3.5 cm).

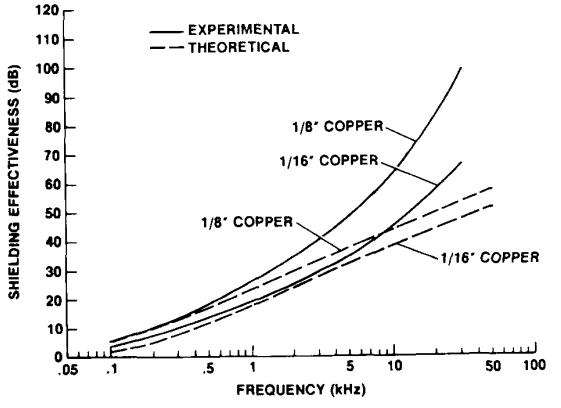


Fig. 3. Comparison of shielding effectiveness S for copper by two techniques (Loop radius = 3.5 cm).

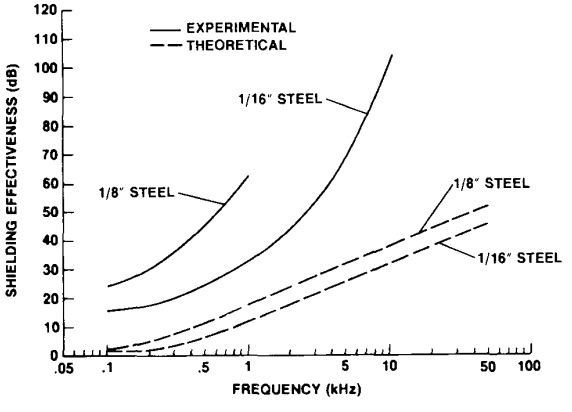


Fig. 4. Comparison of shielding effectiveness S for steel by two techniques (loop radius = 3.5 cm).

Applying the technique of separation of variables by letting $A_\phi = R(r)Z(z)$ and dividing through by $R(r)Z(z)$, we obtain

$$\frac{1}{Z}\frac{\partial^2 Z}{\partial z^2}+k_0^2=\frac{1}{r^2}-\frac{1}{rR}\frac{\partial R}{\partial r}-\frac{1}{R}\frac{\partial^2 R}{\partial r^2}. \tag{14}$$

This must be valid for all values of r and z. Since the right side does not contain z and hence cannot vary with z, so must the sum of terms on the left side be independent of z, and therefore at most be a constant (arbitrarily chosen here to be λ^2). The right side, by similar arguments, must also equal λ^2 for the equality to hold, yielding

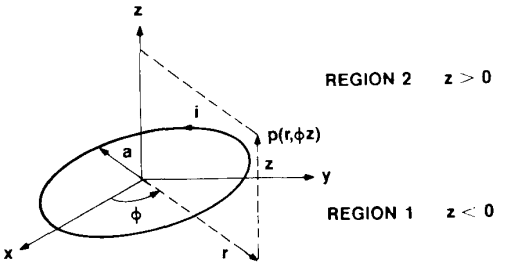


Fig. 5. Loop 1 (field source) in free space.

$$\frac{\partial^2 Z}{\partial z^2}+(k_0^2-\lambda^2)Z=0 \tag{15}$$

$$\frac{\partial^2 R}{\partial r^2}+\frac{1}{r}\frac{\partial R}{\partial r}+\left(\lambda^2-\frac{1}{r^2}\right)R=0. \tag{16}$$

The solution of (15) is

$$Z(z)=Ae^{\tau_0 z}+Be^{-\tau_0 z} \tag{17}$$

where

$$\tau_0=\sqrt{\lambda^2-k_0^2}$$

and

$$k_0^2=\omega^2\mu_0\epsilon_0.$$

The solution of (16) is

$$R(r)=C_1J_1(\lambda r)+C_2Y_1(\lambda r). \tag{18}$$

To satisfy the boundary conditions at $r = 0$ in (18) and at $z = \pm\infty$ in (17), let $C_2 = 0$ and write

$$A_\phi=C_1J_1(\lambda r)\begin{cases}Ae^{\tau_0 z}, & z<0\\ Be^{-\tau_0 z}, & z>0\end{cases}$$

(taking the root of $\lambda^2 - k_0^2$ which has its real part greater than zero).

If the solution is to extend to infinity radially, there are no cylindrical boundaries to impose restrictions on the radial function $R(r)$, and therefore, λ has a continuous, rather than a discrete, spectrum. Thus, the solution involves an integral over all λ[7]:

$$A_\phi^{(1)}=\int_0^\infty B_1(\lambda)J_1(\lambda r)e^{\sqrt{\lambda^2-k_0^2}z}\,d\lambda, \quad z<0 \tag{19}$$

$$A_\phi^{(2)}=\int_0^\infty B_2(\lambda)J_1(\lambda r)e^{-\sqrt{\lambda^2-k_0^2}z}\,d\lambda, \quad z>0. \tag{20}$$

At $z = 0$, $A_\phi^{(1)} = A_\phi^{(2)}$; so $B_1(\lambda) = B_2(\lambda) = B(\lambda)$. Also

$$\hat{n}\times(H_2-H_1)=\text{true surface current} \tag{21}$$

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where $\hat{n} = \pm \hat{l}_z$ and H_2 and H_1 are radial components of magnetic field intensity. Therefore, from (12) and the vector relation $\vec{B} = \nabla \times \vec{A}$

$$\left.\frac{\partial A_\phi^{(1)}}{\partial z}\right|_{z=0-\epsilon} - \left.\frac{\partial A_\phi^{(2)}}{\partial z}\right|_{z=0+\epsilon} = \mu_0 i_a \delta(r-a). \tag{22}$$

Performing the indicated operations on (19) and (20) and letting ϵ approach zero, we obtain

$$2\int_0^\infty B(\lambda')J_1(\lambda' r)\sqrt{\lambda'^2 - k_0^2}\, d\lambda' = \mu_0 i_d \delta(r-a) \tag{23}$$

where λ' replaces λ as a dummy variable. We now multiply through by $J_1(\lambda r) r\, dr$ and integrate

$$2\int_0^\infty J_1(\lambda r) r\, dr \int_0^\infty B(\lambda')\sqrt{\lambda'^2 - k_0^2} J_1(\lambda' r)\, d\lambda = \int_0^\infty \mu_0 i_d \delta(r-a) J_1(\lambda r) r\, dr. \tag{24}$$

We may make use of the Fourier–Bessel integral, which states that, with suitable restrictions on n and $f(x)$[8]

$$f(x) = \int_0^\infty tJ_n(tx)\left[\int_0^\infty \mu f(\mu) J_n(t\mu)\, d\mu\right] dt. \tag{25}$$

If we extract λ' from $\sqrt{\lambda'^2 - k_0^2}$ in (24) and let $\lambda' = \mu$, $r = t$, $x = \lambda$, (25) yields directly

$$B(\lambda) = \frac{\mu_0 a i}{2} \frac{\lambda}{\sqrt{\lambda^2 - k_0^2}} J_1(\lambda a). \tag{26}$$

Thus, we obtain

$$A_\phi^{(1)} = \frac{\mu_0 a i}{2}\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{\tau_0 z}\, d\lambda, \quad z<0 \tag{27}$$

$$A_\phi^{(2)} = \frac{\mu_0 a i}{2}\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{-\tau_0 z}\, d\lambda, \quad z>0. \tag{28}$$

These completely and uniquely describe the vector potential for all space surrounding the loop.

To check the results we use the well-known identity

$$\int_0^\infty e^{-\tau_0 z} J_0(\lambda a)\frac{\lambda}{\tau_0}\, d\lambda = \frac{e^{-jk_0 r}}{r} \tag{29}$$

where

$$r^2 = a^2 + z^2.$$

But

$$d/da[J_0(\lambda a)] = -\lambda J_1(\lambda a).$$

Therefore, taking the derivative of both sides of (29) with respect to a, we obtain

$$\left(jk_0 a + \frac{a}{\sqrt{a^2+z^2}}\right)\frac{e^{-jk_0\sqrt{a^2+z^2}}}{a^2+z^2} = \int_0^\infty \frac{\lambda^2}{\tau_0} e^{-\tau_0 z} J_1(\lambda a)\, d\lambda. \tag{30}$$

However, from (12)

$$B_z = (\nabla \times \vec{A})_z = \frac{\partial A_\phi}{\partial r} + \frac{1}{r} A_\phi = \frac{\mu_0 a i}{2}\int_0^\infty \frac{\lambda^2}{\tau_0} J_1(\lambda a) J_0(\lambda r) e^{-\tau_0 z}\, d\lambda.$$

Therefore, at $r = 0$, we have $J_0(\lambda r) = 1$ for all finite λ and

$$B_z = \frac{\mu_0 a i}{2}\left(jk_0 a + \frac{a}{\sqrt{a^2+z^2}}\right)\frac{e^{-jk_0\sqrt{a^2+z^2}}}{a^2+z^2}. \tag{31}$$

For frequencies below 100 kHz and loop radii less than 5 cm, $k_0 a = \omega a\sqrt{\mu_0\epsilon_0}$ is less than 10^{-4}. For $z < 9.9a$, $a/\sqrt{a^2+z^2}$ is greater than 10^{-1}. Therefore, for the range of variables $f < 10^5$ Hz, $a < 5$ cm, and $z < 10a$, we see that the product $k_0\sqrt{a^2+z^2}$ is less than 10^{-3}. Hence, we can write to a good approximation

$$B_z = \frac{\mu_0 a^2 i}{2(a^2+z^2)^{3/2}} \tag{32}$$

which agrees with the result for magnetostatic flux density along the axis of a loop[9].

Introducing the conducting plate as shown in Fig. 6, we have $A_\phi = A_\phi^{\text{impressed}} + A_\phi^{\text{induced}}$. The eddy current density within the plate has ϕ symmetry; therefore the induced vector potential will be the same formally as the impressed vector potential, modified at most by some function of λ

$$A_\phi^{(1)} = \frac{\mu_0 a i}{2}\left\{\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{-\tau_0|z|}\, d\lambda + \int_0^\infty C(\lambda)\frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{\tau_0 z}\, d\lambda\right\} \tag{33}$$

$$A_\phi^{(2)} = \frac{\mu a i}{2}\left\{\int_0^\infty D(\lambda)\frac{\lambda}{\tau} J_1(\lambda a) J_1(\lambda r) e^{-\tau z}\, d\lambda + \int_0^\infty E(\lambda)\frac{\lambda}{\tau} J_1(\lambda a) J_1(\lambda r) e^{\tau z}\, d\lambda\right\} \tag{34}$$

$$A_\phi^{(3)} = \frac{\mu_0 a i}{2}\int_0^\infty F(\lambda)\frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{-\tau_0 z}\, d\lambda \tag{35}$$

where

$$\tau = \sqrt{\lambda^2 - k^2}$$

and

$$k^2 = j\omega\mu\sigma.$$

At $z = b$

$$A_\phi^{(1)} = A_\phi^{(2)}$$

$$\frac{1}{\mu_0}\frac{\partial A_\phi^{(1)}}{\partial z} = \frac{1}{\mu}\frac{\partial A_\phi^{(2)}}{\partial z}. \tag{36}$$

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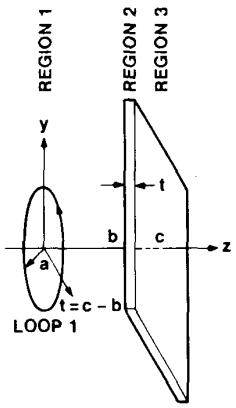


Fig. 6. Loop 1 and infinite shield plate of relative conductivity σ_r, relative permeability μ_r.

At $z = c$

$$A_\phi^{(2)} = A_\phi^{(3)}$$

$$\frac{1}{\mu}\frac{\partial A_\phi^{(2)}}{\partial z} = \frac{1}{\mu_0}\frac{\partial A_\phi^{(3)}}{\partial z}. \tag{37}$$

Performing the indicated operations, letting $K = \tau/\tau_0$, $\mu_r = \mu/\mu_0$, and $t = c - b$, we obtain

$$\begin{array}{llll} 1)\ Ke^{\tau_0 b}C(\lambda) & -\mu_r e^{-\tau b}D(\lambda) & -\mu_r e^{\tau b}E(\lambda) & +0 \qquad\qquad = -Ke^{-\tau_0 b} \\ 2)\ e^{\tau_0 b}C(\lambda) & +e^{-\tau b}D(\lambda) & -e^{\tau b}E(\lambda) & +0 \qquad\qquad = e^{-\tau_0 b} \\ 3)\ 0 & +\mu_r e^{-\tau c}D(\lambda) & +\mu_r e^{\tau c}E(\lambda) & -KF(\lambda)e^{-\tau_0 c} = 0 \\ 4)\ 0 & -e^{-\tau c}D(\lambda) & +e^{\tau c}E(\lambda) & +\quad e^{-\tau_0 c}F(\lambda) = 0 \end{array}$$

$$\Delta = e^{-\tau_0(c-b)}\begin{vmatrix} K & -\mu_r e^{-\tau b} & -\mu_r e^{\tau b} & 0 \\ 1 & e^{-\tau b} & e^{+\tau b} & 0 \\ 0 & \mu_r e^{-\tau c} & \mu_r e^{\tau c} & -K \\ 0 & -e^{-\tau c} & e^{\tau c} & 1 \end{vmatrix}$$

$$\Delta = Ce^{t(\tau-\tau_0)}$$

where

$$C = [K+\mu_r)^2 - (K-\mu_r)^2 e^{-2t\tau}]^{-1}. \tag{38}$$

Solving similar determinants to find $C(\lambda)$, $D(\lambda)$, $E(\lambda)$, and $F(\lambda)$, we have

$$C(\lambda) = Ce^{-2b\tau_0}\{(K^2-\mu_r^2)(e^{-2t\tau}-1)\} \tag{39}$$

$$D(\lambda) = 2CKe^{b(\tau-\tau_0)}(K+\mu_r) \tag{40}$$

$$E(\lambda) = 2CKe^{-(t+c)\tau - b\tau_0}(K-\mu_r) \tag{41}$$

$$F(\lambda) = 4CK\mu_r e^{-t(\tau-\tau_0)}. \tag{42}$$

Substituting (39)–(42) into (33)–(35), we get

$$A^{\phi(1)} = \frac{\mu_0 ai}{2}\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)[e^{-\tau_0|z|} + Ce^{\tau_0(z-2b)}(K^2-\mu_r^2)(e^{-2t\tau}-1)]\,d\lambda \tag{43}$$

$$A_\phi^{(2)} = \mu ai\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)Ce^{b(\tau-\tau_0)} \cdot [e^{-\tau z}(K+\mu_r) + e^{\tau(z-2c)}(K-\mu_r)]\,d\lambda \tag{44}$$

$$A_\phi^{(3)} = 2\mu ai\int_0^\infty \frac{C\lambda\tau}{\tau_0^2} J_1(\lambda a)J_1(\lambda r)e^{-\tau_0 z - t(\tau-\tau_0)}\,d\lambda \tag{45}$$

where C is given by (38). These results are in agreement with those of Smythe[10] if we let $\omega = 0$.

The conditions set forth in (10) still hold when the shield is introduced; therefore, by the use of (12)

$$\vec{B}^{(1)} = \nabla \times \vec{A}^{(1)}$$

$$= \frac{\mu_0 ai}{2}\left\{\hat{l}_r\int_0^\infty \lambda J_1(\lambda a)J_1(\lambda r) \cdot [e^{-\tau_0|z|} - Ce^{\tau_0(z-2b)}(K^2-\mu_r^2)(e^{-2t\tau}-1)]\,d\lambda + \hat{l}_z\int_0^\infty \frac{\lambda^2}{\tau_0} J_0(\lambda r)J_1(\lambda a) \cdot [e^{-\tau_0|z|} + Ce^{\tau_0(z-2b)}(K^2-\mu_r^2)(e^{-2t\tau}-1)]\,d\lambda\right\} \tag{46}$$

$$\vec{E}^{(1)} = -j\omega\vec{A}^{(1)} + \frac{\nabla(\nabla\cdot\vec{A}^{(1)})}{j\omega\mu\epsilon} = -j\omega\vec{A}^{(1)}\ \text{[from (11)]} \tag{47}$$

$$\vec{E}^{(1)} = -\hat{l}_\phi\frac{j\omega\mu_0 ai}{2}\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r) \cdot [e^{-\tau_0|z|} + Ce^{\tau_0(z-2b)}(K^2-\mu_r^2)(e^{-2t\tau}-1)]\,d\lambda \tag{48}$$

$$\vec{B}^{(2)} = \mu ai\left\{\hat{l}_r\int_0^\infty \frac{\tau\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)Ce^{b(\tau-\tau_0)} \cdot [e^{-\tau z}(K+\mu_r) - e^{r(z-2c)}(K-\mu_r)]\,d\lambda + \hat{l}_z\int_0^\infty \frac{\lambda^2}{\tau_0} J_1(\lambda a)J_0(\lambda r)Ce^{b(\tau-\tau_0)} \cdot [e^{-\tau z}(K+\mu_r) + e^{\tau(z-2c)}(K-\mu_r)]\,d\lambda\right\} \tag{49}$$

$$\vec{E}^{(2)} = -j\omega\mu ai\hat{l}_\phi\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)Ce^{b(\tau-\tau_0)} \cdot [e^{-\tau z}(K+\mu_r) + e^{\tau(z-2c)}(K-\mu_r)]\,d\lambda \tag{50}$$

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$$\vec{B}^{(3)} = 2\mu a i \left\{ \hat{l}_r \int_0^\infty \frac{C\lambda\tau}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{-\tau_0 z - t(\tau-\tau_0)}\, d\lambda \cdot \hat{l}_z \int_0^\infty \frac{C\lambda^2\tau}{\tau_0^2} J_1(\lambda a) J_0(\lambda r) e^{-\tau_0 z - t(\tau-\tau_0)}\, d\lambda \right\} \quad (51)$$

$$\vec{E}^{(3)} = -2j\omega\mu a i \hat{l}_\phi \int_0^\infty \frac{C\lambda\tau}{\tau_0^2} J_1(\lambda a) J_1(\lambda r) e^{-\tau_0 z - t(\tau-\tau_0)}\, d\lambda. \quad (52)$$

To obtain the free space flux density, we use (28) to obtain

$$\vec{B} = \nabla \times \vec{A} = \frac{\mu_0 a i}{2} \left\{ \hat{l}_r \int_0^\infty \lambda J_1(\lambda a) J_1(\lambda r) e^{-\tau_0 z}\, d\lambda + \hat{l}_z \int_0^\infty \frac{\lambda^2}{\tau_0} J_1(\lambda a) J_0(\lambda r) e^{-\tau_0 z}\, d\lambda \right\}.$$

Finally, we are in a position to write an expression for the shielding effectiveness of the infinite plate. For the loop orientation being considered, it is consistent with (7) to write

$$S = 20 \log_{10} \left| \frac{B_z}{B_z^{(3)}} \right| \quad (53)$$

where B_z is the z-directed flux density at the same point on the z axis, but with the shield removed. Writing it out, we have

$$S = 20 \log_{10} \frac{1}{4\mu_r} \left| \frac{\int_0^\infty \frac{\lambda^2}{\tau_0} J_1(\lambda a) e^{-\tau_0 z}\, d\lambda}{\int_0^\infty \frac{C\lambda^2\tau}{\tau_0^2} J_1(\lambda a) e^{-\tau_0 z - t(\tau-\tau_0)}\, d\lambda} \right|. \quad (54)$$

We could use (32) to simplify (54).

$$S = -20 \log_{10} \frac{4\mu_r}{a} (a^2+z^2)^{3/2} \cdot \left| \int_0^\infty \frac{C\lambda^2\tau}{\tau_0^2} J_1(\lambda a) e^{-\tau_0 z - t(\tau-\tau_0)}\, d\lambda \right|. \quad (55)$$

Note the independence of S with source to shield spacing $b(=r_1)$, a result which will be shown to be consistent with experimental results to be reported on herein.

Ryan [14] has solved (54) using numerical integration. His results agree very favorably with experimental results contained herein. Using numerical analysis Bannister [12] has subsequently shown the equivalence of (54) with results from Schelkunoff's [4] transmission theory of shielding. Since those results are much more manageable when properly applied, the transmission theory will next be reviewed.

Schelkunoff considers a transmission line of length l inserted between another line of propagation constant Γ_1 and characteristic impedance K_1, and a load impedance Z. He shows[3] that the transmission coefficient across the inserted uniform line of propagation constant Γ_2 and characteristic impedance K_2 is

$$T' = p'(1 - q' e^{-2\Gamma_2 l})^{-4} e^{-\Gamma_2 l} \quad (56)$$

where

$$p' = \frac{4K_1K_2}{(K_1+K_2)(Z+K_2)}$$

and

$$q' = \frac{(K_2-K_1)(K_2-Z)}{(K_1+K_2)(Z+K_2)}.$$

Since waves within a plane sheet of metal have field components almost completely in the plane of the shield[11] it is logical to make an analogy with a practical transmission line that can be almost entirely TEM. If we denote the wave impedance on either side of the shield by $Z_{\omega 1}$ and $Z_{\omega 3}$, and if we let $Z_{\omega 1}$ equal $Z_{\omega 3}$ (thus neglecting wave divergence in traversing our physically thin shield) and define the wave impedance and propagation constant within the shield to be η and $\bar{\sigma}$, respectively, one can argue by analogy that the transmission coefficient across the inserted homogeneous shield of thickness t is

$$T = p(1 - qe^{-2\sigma t})^{-1} e^{-\sigma t} \quad (57)$$

where

$$p = \frac{4k}{(k+1)^2}, \quad q = \frac{(k-1)^2}{(k+1)^2}$$

and k is the ratio η/Z_ω or Z_ω/η, Z_ω being the incident wave impedance normal to the shield. Then the insertion loss in decibels between two corresponding points just outside the plate would be for the transverse field components

$$\begin{aligned} S &= -20 \log_{10} |T| \\ &= 20 \log_{10} \left| \frac{1}{p} \right| - 20 \left(\frac{1}{2.30259} \ln |e^{-\bar{\sigma} t}| \right) \\ &\quad + 20 \log_{10} |1 - qe^{-2\bar{\sigma} t}| \\ &= R + A + B \text{ in decibels.} \end{aligned} \quad (58)$$

Thus, in general, we must calculate

$$\begin{aligned} R &= 20 \log_{10} \frac{|k+1|^2}{4|k|} \text{ in decibels} \\ &= 10 \log_{10} [0.00286k^2 + 0.0378k \\ &\quad + 0.25 + 0.826k^{-1} + 1.365k^{-2}] \end{aligned} \quad (59)$$

where $k = \rho\sqrt{f\mu_r\sigma_r}$ and

ρ = distance of magnetic dipole from shield (in centimeters)

$$A = -8.686 \ln |e^{-(1+j)\alpha t}| = 8.686\, \alpha t \text{ in decibels} \quad (60)$$

$$B \Rightarrow 20 \log_{10} |1 - qe^{-2\bar{\sigma} t}| \text{ in decibels} \quad (61)$$

[3] Schelkunoff[4] pp. 223-225.

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where

$$\bar{\sigma}=\sqrt{j\omega\mu(\sigma+j\omega\epsilon)} \text{ and } \eta=\sqrt{\frac{j\omega\mu}{\sigma+j\omega\epsilon}}.$$

For any good shield in the frequency range of interest $\sigma/\omega\epsilon \gg 1$. Therefore, by choosing the root of j that lies in the first quadrant (since this choice is arbitrarily but consistently made in the literature), these become

$$\bar{\sigma}=(1+j)\sqrt{\pi f\mu\sigma} \equiv (1+j)\alpha,\ \eta=(1+j)\sqrt{\pi f\mu/\sigma}.$$

Thus, R represents the reflection loss due to the initial reflection at both interfaces of the shield, B accounts for all higher order reflections not contained in R, and A is the attenuation of the field components in penetrating the shield once. The most difficult component to calculate is B, but fortunately, as A gets large, B becomes negligible. Usually B is neglected for values of A in excess of 15 dB. The reader is referred to Schelkunoff for detailed discussion of this theory.

Some remarks should be made concerning the impedance of the wave which is incident onto the shield. An exact description of the fields in the immediate vicinity of the source (loop 1) would involve expressions containing complete elliptic integrals of the first and second kind. One would be hard put to apply meaningfully the resulting expression for the incident wave impedance Z_ω in (59) and (61). There is reason to believe that the fields about loop 1 do not differ markedly from those from a similarly oriented elemental magnetic dipole. Forming the ratio of E_θ to H_ϕ in the elemental dipole formulas[4] and considering the near field where $2\pi\rho/\lambda \ll 1$, being the radial distance from the center of the dipole, one obtains for the radial wave impedance at ρ

$$Z_\omega=j\omega\mu_0\rho. \tag{62}$$

We shall use this approximation for Z_ω of loop 1.

In our problem one would not, in general, expect (58) to predict the correct insertion loss created by placing the shield arbitrarily between the two loops. Consider the case when A exceeds 15 dB, thereby making B negligible. The analyses leading to (7) and (54) indicate that the insertion loss S is *not* a function of the shield position between the two fixed loops. By the theory just outlined, A would be easily calculable by knowing the intrinsic properties of the shield and the source frequency of oscillation. But by (62) it is not correct to assume that R would be independent of shield position. The incident wave impedance, and hence R, would vary as one proceeded radially away from the z-axis along the source-side of the shield. It would seem that this variation would generally become larger as the loop 1 to shield spacing b decreased. But at loop 2, the shielded point of observation, the integrated effect of these circles of constant R would have to be independent of the shield position if the theories are to be consistent. We should, however, be able to virtually displace the shield between the source coil and the point of observation for calculation purposes. Imagine the shield to be displaced to coincide with loop 2. Thus, to evaluate (58), we would replace ρ by the loop-to-loop spacing z in (62). Since the longitudinal field components are reduced by the same amount as the transverse components when the shield is present, one might expect (58) to predict the insertion loss created by the introduction of the shield between the two loops. The accuracy would be expected to increase as the loop-to-loop spacing increased.

[4] Schelkunoff[4] p. 163.

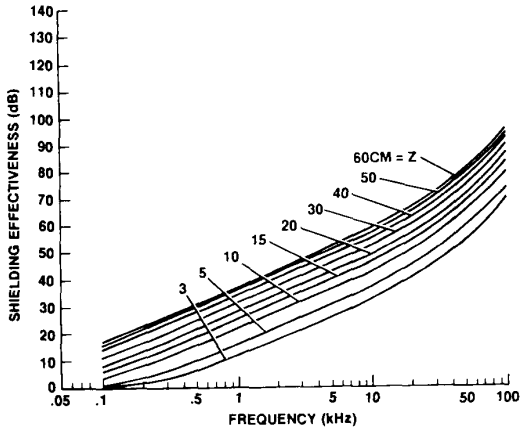


Fig. 7. Theoretical shielding effectiveness S versus frequency by Schelkunoff's $S = R + A + B$ for 1/16-in aluminum.

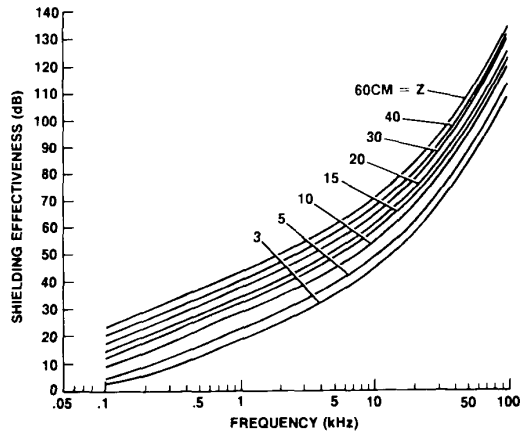


Fig. 8. Theoretical shielding effectiveness S versus frequency by Schelkunoff's $S = R + A + B$ for 1/16-in copper.

Figs. 7–9, which are graphs of shielding effectiveness S versus frequency, were computer derived from (58). Results from 1/8-in thick aluminum, copper, and steel are not included herein due to space limitations. See [15] for complete data. Three materials were chosen for analysis: one having relative permeability and relative conductivity equal to unity (annealed copper), one having μ_r but not σ_r equal to unity (5052 aluminum alloy $\sigma_r = 0.35$), and one having neither μ_r nor σ_r equal to unity (steel, $\sigma_r = 0.17$, $\mu_r = 112$). Each shield was evaluated for two thicknesses, $t = 1/16$ in and $t = 1/8$ in (corresponding to the six plates to be used later for experimental laboratory determination of shielding effectiveness.

Comparing these data with the corresponding data obtained experimentally (Figs. 11–12), it appears that agreement is not as predicted by our proposed virtual displacement technique;

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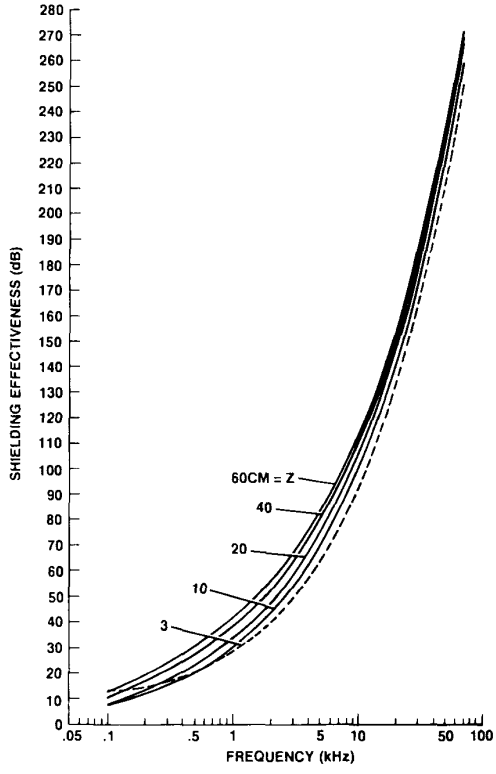


Fig. 9. Theoretical shielding effectiveness S versus frequency by Schelkunoff's $S = R + A + B$ for 1/16-in steel.

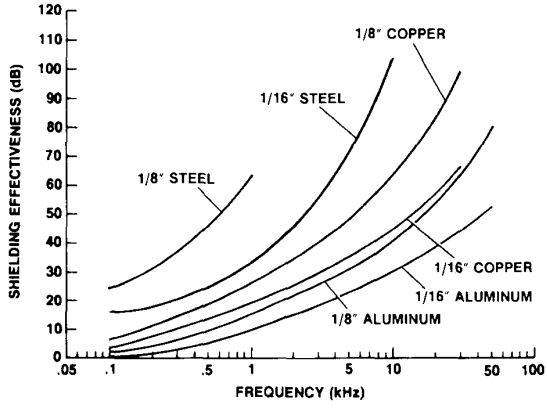


Fig. 10. Experimental insertion loss S versus frequency for the six shields considered (all curves for loop 1 radius = 3.5 cm, loop spacing z = 3 cm.)

differences generally tend to increase with increase in loop spacing. *However, if one considers the plate to be virtually displaced such that we have (see Fig. 1)*

$$\frac{r_1}{r_1+r_2}=\frac{b}{z}=\frac{1}{3} \tag{63}$$

careful comparison of corresponding curves suggests that (58) is useful in predicting within 3 dB the total insertion loss of the aluminum and copper shields. This relationship has been verified by Bannister's numerical analysis of 54 [12].

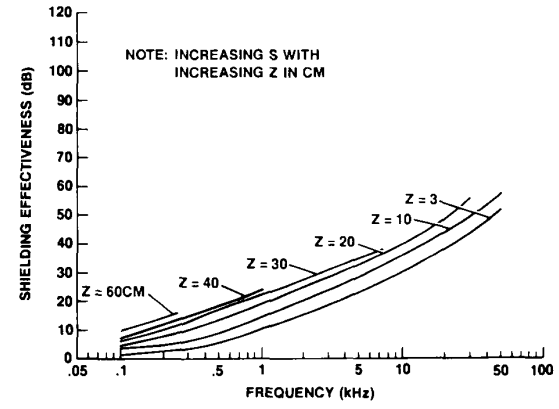


Fig. 11. Experimental shielding effectiveness S versus frequency for 1/16-in aluminum.

In Figs. 7–9 and 11–12 the parameter z represents the loop 1 to point of observation axial distance.

III. Experimental Data

Last in our discussion is the experimental solution of the problem. A test arrangement that allowed the two coaxial loops to be kept at a constant separation z while a 4-ft by 4-ft shield was moved between them from one extreme to the other was constructed. Loops of 3.5-cm radius and 7 turns were used for loops 1 and 2 of Fig. 1. The independence of S and the loop 1 to shielded distance b predicted in (7) and (54) has been demonstrated experimentally for different shielding materials and thicknesses. This provided the experimenter with a means of detecting data contaminated by end effects. When a variation of S with shield position between the two loops was observed, the data were not taken as a true representation of the shielding effectiveness of the shield being tested. It was consistently observed that end effects seem to become more and more pronounced with increasing frequency, shielding effectiveness of the shield, and loop spacing z; i.e., for the more effective shields, end effects occurred at much lower frequencies for the same loop spacing. Figs. 11–12 graphically illustrate the shielding effect versus frequency curves for two of the six plates tested. Space limitations preclude inclusion of all experimental data obtained. Complete data for all samples tested is contained in [15]. Less meaningful shielding effectiveness data are obtainable with the more effective shields as loop spacing or frequency is increased. An explanation of this phenomenon follows:

1) The leakage fields are assumed to be independent of the frequency. They should be relatively independent of loop spacing z since they are mainly a function of the leakage path length around the shield. But the leakage voltage V_L induced in loop 2 will be a linear increasing function of frequency if the field and loop positions are held constant.

2) From Fig. 10 the direct coupling voltage V_D to loop 2 goes down (between 1 and 50 kHz for the materials tested) by a complex function of frequency. Therefore, one can see that the ratio V_L/V_D is an increasing function of frequency, shielding effectiveness, and loop spacing z. It is this ratio which should be kept small when shielding effect measurements are being conducted.

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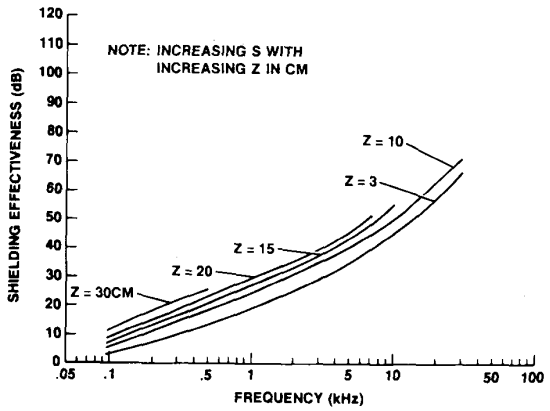


Fig. 12. Experimental shielding effectiveness S versus frequency for 1/16-in copper.

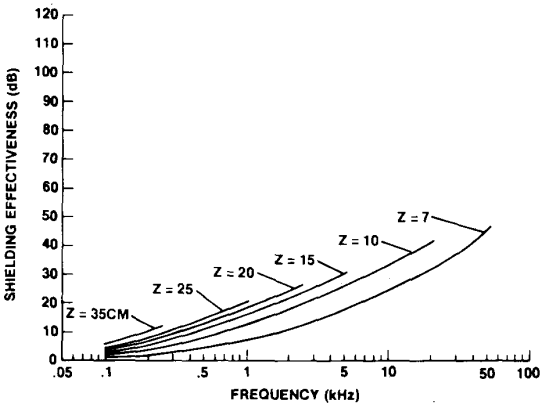


Fig. 13. Experimental shielding effectiveness S versus frequency for 1/16-in aluminum. (Loop axes parallel to shield loop center-to-center distance z in centimeters.)

Fig. 13 represents shielding data obtained with the two loop axes parallel to the plane of the shield interfaces. From the data obtained by using the 1/16-in aluminum shield, the shielding does not differ greatly from that observed with the original loop orientation, especially for the larger loop separations. This strongly supports the assumption that the wave impedance can be calculated on the basis of an elemental magnetic dipole at radial distances less than would seem justifiable.

Fig. 10 summarizes the shielding effectivenesses of 1/16-in thick aluminum, copper and steel for loop spacing $Z = 3$ cm and source (loop 1) radius = 3.5 cm.

IV. Conclusion

Figs. 2 and 3 show that the experimental and theoretical results obtained by the first analytical method are within 3 dB for values of shielding effectiveness below 25 dB for the aluminum and copper sheets. Agreement seems to get better as the frequency is reduced. Inspection of Fig. 4 implies that (7) is not accurate for shielding effectiveness determination of sheets with relative permeability much different from unity. Thus, the neglected quantities (magnetic permeability and axial variation of eddy currents within the sheet) tend to limit the usefulness of our first method for solving this problem. Bannister [13] has shown that (7) is a particular case of the exact solution to the vector wave equation (54) contained herein.

The integral expression for shielding effectiveness contained herein proves the independence of shielding effectiveness with source to shield spacing. Ryan [14] has used numerical integration of (55) to show results in excellent agreement with experimental data contained herein. Bannister [12] has used numerical analysis to show equivalence of (55) and (59) if and only if the constraints of (63) are imposed.

The third analytical technique is adapted to our problem from Schelkunoff's well-known transmission theory of shielding. By direct comparison of the theoretical data (which are computer-derived from the results of this theory) with the experimentally obtained data, we were able to determine the usefulness of our third analytical technique. Agreement between experimental and theoretical values of shielding effectiveness for aluminum and copper was attained within 3 dB provided the shield was placed (for calculation) a distance from the source equal to 1/3 of the loop spacing z (63).

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